MASTER OF SCIENCE
IN ENGINEERING

Statistical Signal Processing and Modelling (Machine Learning)

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Course planning

Week No. (MSE)	Date	Lecturer	Topic	Sub Topic
1	16.02.2026	M.Tognolini	Chap 3 : Discrete-Time Random Process	intro + Generate random variable
2	23.02.2026	M.Tognolini	Chap 3 : Discrete-Time Random Process	random process, autocorrelation
3	02.03.2026	M.Tognolini	Chap 3 : Discrete-Time Random Process	randon process Power spectrum and filtered signals
4	09.03.2026	M.Tognolini	Chap 4 : Signal Modeling	padé, prony
5	16.03.2026	M.Tognolini	Chap 4 : Signal Modeling	stochastic model
6	23.03.2026	M.Tognolini	Chap 4 : Signal Modeling	optionel
7	30.03.2026	M.Tognolini	Chap XI : Wavelet introduction	
	06.04.2026	Easter		
8	13.04.2026	D.Lavanchy	Chap 7 : Optimum filters	FIR Wiener filter
9	20.04.2026	D.Lavanchy	Chap 7 : Optimum filters	Kalman filter
10	27.04.2026	D.Lavanchy	Chap 9 : Adaptive filtering	FIR adaptive filters
11	04.05.2026	D.Lavanchy	Chap 9 : Adaptive filtering	Recursive Least squares
12	11.05.2026	D.Lavanchy	Chap 9 : Adaptive filtering	Application pratique
13	18.05.2026	D.Lavanchy	Chap 8 : Spectrum estimation	Non-parametric approach
	25.05.2026	Lun Pentecote		
14	01.06.2026	D.Lavanchy	Chap 8 : Spectrum estimation	parametric approach
	08.06.2026	D.Lavanchy, M.T	Q&R (compensation fériés)	
	15.06.2026	Exam Preparation		
	22.06.2026	EXAMS		
	29.06.2026	EXAMS		

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Module organization

- **Lectures :** 1 periods each Monday morning.
- **Exercises:** 2 period each Monday after lecture.
- **Personal work at home:** about 3 periods per week are needed to master the theory. Finish Matlab exercices and complete the notes on the slides and for more explanation read the related chapter in the textbook (Hayes or/and Kocher).
- **Computer problems:** The computer problems during the exercise session and for the homework are based on PYTHON JPYNB.
- **Exams**: Individual oral exam on the lab done during the semester.

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TextBook and Labs

- For this Master Module you need the text book (TB): **Statistical Digital Signal Processing and Modeling, Monson H. HAYES Ed. Wiley** (Teams: General/ Support de cours)
- A PDF document edited by M.Kocher & M.Tognolini is also useful for understanding methods and exercises. (Teams: General/ Support de cours)
- Exercises for the module are based on this textbook.
- Matlab Labs are on GitHub classroom, a link for week assignment is given on Teams each week session.
- Slides of the module contains a part of the matter of the book. (Teams: General/ Presentations)
- The final exams are based on the slides and the lab exercises.

Statistical signal applications

- Speech synthesis: The model of the vocal system permit to do a synthesis of the speech.
- Time series prediction models for economy: Prediction of the shares values….
- Time of flight detection for sensors with electromagnetic waves (RADAR) or laser light (LIDAR) or ultrasound for liquids (SONAR).
- Tracking object or movement in video
- Measuring baby electrocardiogram disturbed by mother electrocardiogram (mother)
- …